

Functional guilds of seaweed degradation & detoxification by algivorous fish gut microbiota

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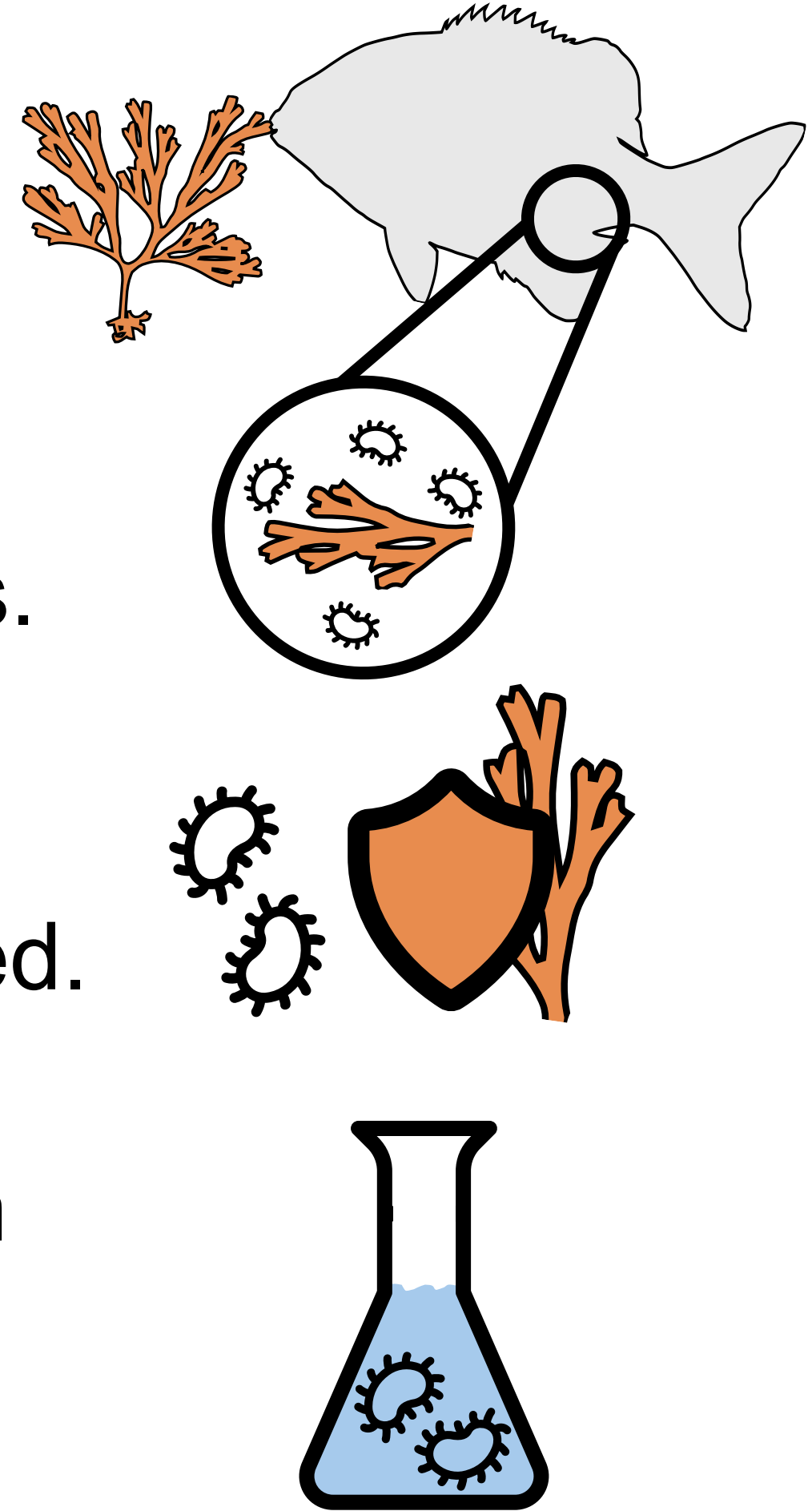
How do gut microbes support algivory?

Kyphosus cinerascens is an *algivorous* (seaweed-eating) tropical fish.

Gut microbiota in *K. cinerascens* use enzymes to break down seaweed sugars.

Seaweed release defensive phenols that inhibit bacterial enzymes unless detoxified.

We can grow *Kyphosus* gut microbiota in artificial microcosms to study how they digest seaweed biomass over time.



How are the most abundant *Kyphosus* gut microbiota involved in this reaction cascade?

Gammaproteobacteria

Bacillota

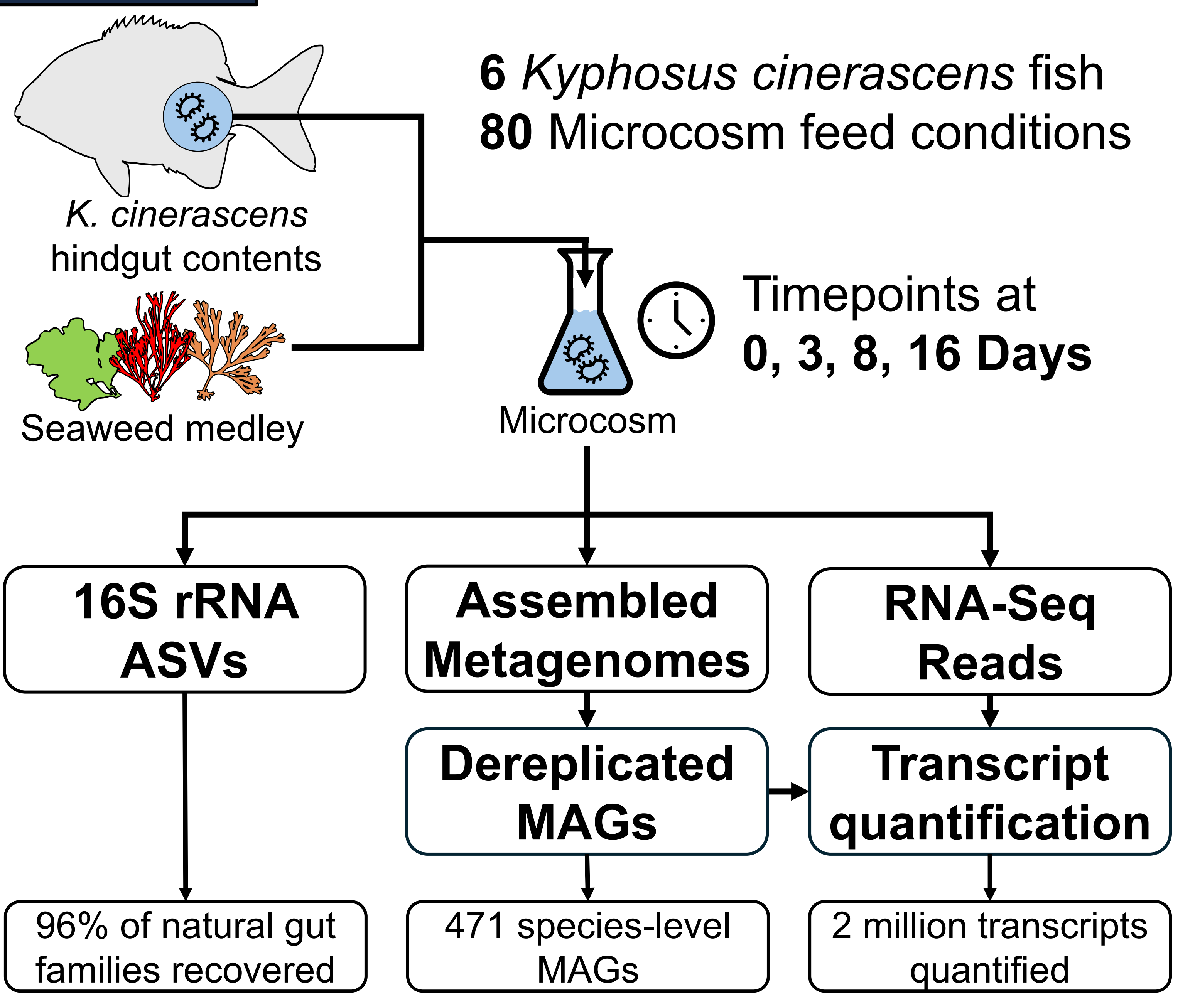
Bacteroidota

Verrucomicrobiota

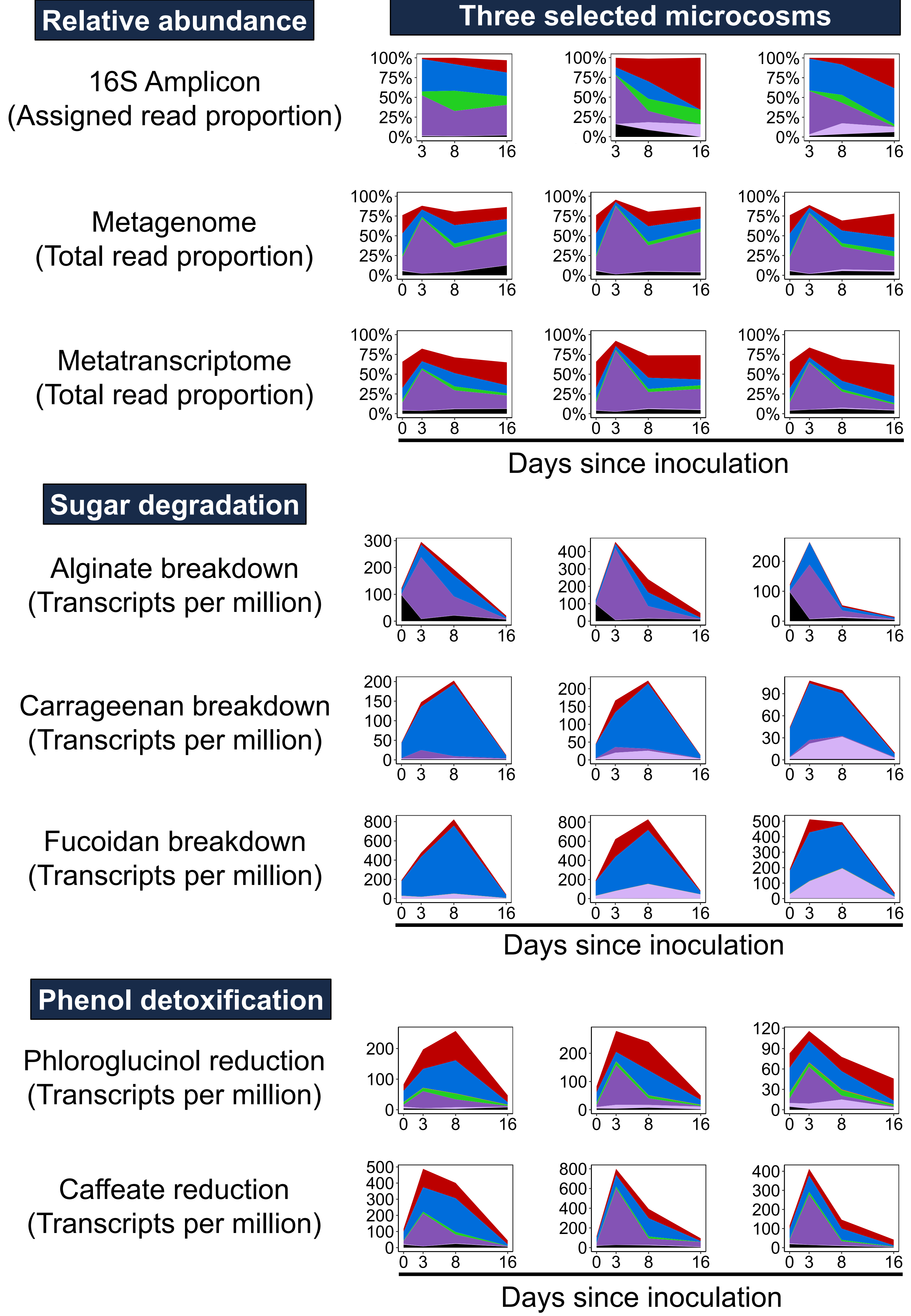
Desulfobacterota

Other Taxa

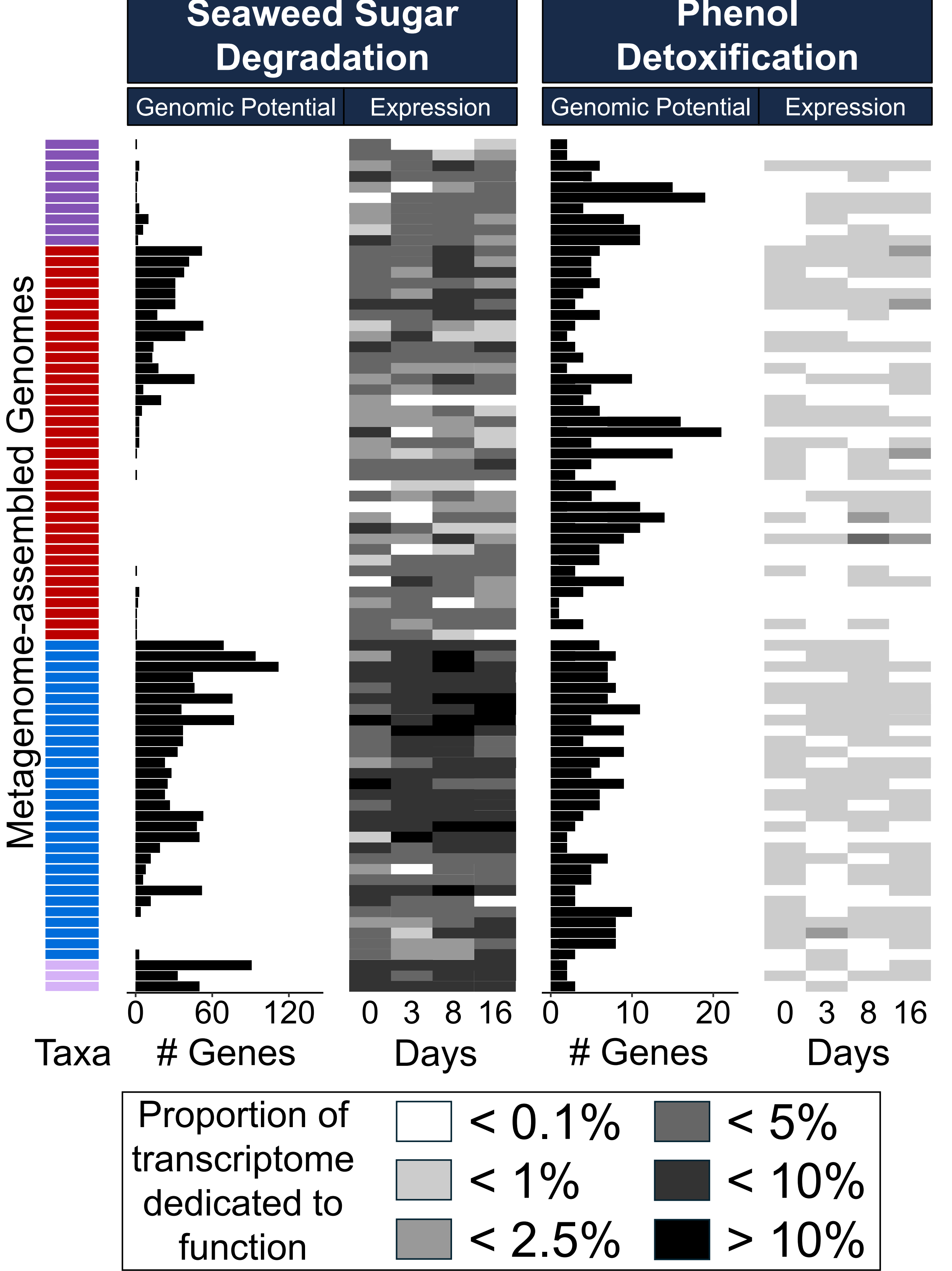
Methods



Temporal succession in microcosms



Divergent niches in related taxa



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Key takeaway: temporal & functional guilds of core microbiota

- Gammaproteobacteria**: Early alginate degrader; Early detoxifier
- Bacillota**: Early alginate degrader; Mid-late generalist; Mid-late detoxifier; Transient scavenger
- Bacteroidota**: Mid-late generalist; Late detoxifier
- Verrucomicrobiota**: Late sulfated polysaccharide degrader
- Desulfobacterota**: Constitutive detoxifier & sulfate reducer